Submission B solution to Problem 6

Overview / Strategy

The proof has two main parts. First, a vertex of weight 1 is automatically irreducible, because a decomposition $u = p + q$ with $p \cdot q \geq 0$ would force $1 = u^2 \geq p^2 + q^2 \geq 2$. Thus, after this easy case, we may assume that no vertex has weight 1.

In that remaining case, let b be the unique vertex with $w(b) < d_T(b)$. We prove that b itself is irreducible. The key input is a rooted-branch estimate for each component of $T - \{b\}$. That estimate is proved by induction using a Schur-complement calculation and a parity argument for the inverse Gram matrix. Combining the branch estimates with the Schur complement at b, any hypothetical decomposition $b = x + (b - x)$ with $x \cdot (b - x) \geq 0$ is forced to have $x = 0$ or $x = b$, contradicting the requirement that both summands be nonzero.

Statement and notation

Let T be a finite tree with vertex set V, edge set E, and weight function $w : V \to \mathbb{Z}$. We write $d_T(v)$ for the degree of a vertex v in T. Let $L(T)$ be the free abelian group generated by V. We identify a vertex with the corresponding basis element of $L(T)$. The bilinear form on $L(T)$ is the symmetric form determined on generators by

$$u \cdot v = \begin{cases} w(u), & u = v, \\ -1, & \{u, v\} \in E, \\ 0, & u \neq v \text{ and } \{u, v\} \notin E. \end{cases}$$

For $x \in L(T)$, write $x^2 = x \cdot x$. The form is called positive definite if its extension to $L(T) \otimes_{\mathbb{Z}} \mathbb{R}$ is positive definite. Since the Gram matrix has integer entries, if the form is positive definite, then every nonzero $z \in L(T)$ satisfies $z^2 \in \mathbb{Z}_{>0}$.

A nonzero lattice element $y \in L(T)$ is called *irreducible* if there do not exist nonzero $a, c \in L(T)$ such that

$$y = a + c \qquad \text{and} \qquad a \cdot c \geq 0.$$

Theorem 1 (Original problem)**.** *Let T be a finite weighted tree with vertex set V, edge set E, and weight function $w : V \to \mathbb{Z}$. Equip $L(T)$ with the symmetric bilinear form above.*

Suppose this form is positive definite, and suppose there is exactly one vertex $b \in V$ such that

$$w(b) < d_T(b).$$

Then T contains a vertex u such that u is irreducible in $L(T)$.

For a connected subgraph or subtree C, write $d_C(u)$ for the degree of u inside C. The group $L(C)$ is the free abelian group on the vertices of C, equipped with the restricted form. If $x \in L(C)$, write x_u for the coefficient of the vertex u in x.

Auxiliary lemmas

Lemma 2 (A weight-one vertex is irreducible)**.** *Assume the form on $L(T)$ is positive definite. If $u \in V$ satisfies $w(u) = 1$, then u is irreducible.*

Proof. Suppose, for contradiction, that

$$u = p + q$$

where $p, q \in L(T)$ are nonzero and $p \cdot q \geq 0$. Since the form is positive definite and integral, $p^2, q^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = u^2 = (p+q)^2 = p^2 + q^2 + 2p \cdot q \geq 1 + 1 + 0 = 2,$$

which is impossible. Therefore no such decomposition exists, and u is irreducible. $\square$

Lemma 3 (Schur complement calculation)**.** *Let*

$$M = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix},$$

where B is symmetric positive definite. Set

$$S = a - \gamma^T B^{-1} \gamma.$$

Then M is positive definite if and only if $S > 0$. When $S > 0$,

$$M^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}\gamma^T B^{-1} \\ -B^{-1}\gamma S^{-1} & B^{-1} + B^{-1}\gamma S^{-1}\gamma^T B^{-1} \end{pmatrix}.$$

Proof. We have the factorization

$$M = \begin{pmatrix} 1 & \gamma^T B^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ B^{-1}\gamma & I \end{pmatrix}.$$

Indeed, multiplying the last two factors gives

$$\begin{pmatrix} S & 0 \\ \gamma & B \end{pmatrix},$$

and multiplying by the first factor gives

$$\begin{pmatrix} S+\gamma^T B^{-1}\gamma & \gamma^T \\ \gamma & B \end{pmatrix} = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix}.$$

The first and third factors are transposes of one another and are invertible. Congruence by an invertible matrix preserves positive definiteness, so M is positive definite exactly when the middle block diagonal matrix is positive definite. Since B is already positive definite, this is equivalent to $S > 0$.

When $S > 0$, invert the displayed factorization:

$$M^{-1} = \begin{pmatrix} 1 & 0 \\ -B^{-1}\gamma & I \end{pmatrix} \begin{pmatrix} S^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\gamma^T B^{-1} \\ 0 & I \end{pmatrix}.$$

Multiplying these three factors gives exactly the stated block formula. □

Lemma 4 (Rooted branch estimate)**.** *Let C be a finite weighted tree rooted at a vertex r. Let A be its Gram matrix, and assume A is positive definite. Suppose*

$$w(r) \geq d_C(r) + 1, \qquad w(u) \geq d_C(u) \quad \textit{for every } u \neq r.$$

Let $G = A^{-1}$, and define

$$\alpha = (A^{-1})_{rr} = e_r^T G e_r,$$

where e_r is the coordinate vector of r. Then $0 < \alpha \leq 1$, and for every $x \in L(C)$, if $c = x_r$, then for every $k \in \mathbb{Z}$,

$$(2k-1)c - x^2 \leq k(k-1)\alpha.$$

Moreover, if $k = 0$ or $k = 1$, and equality holds with $x \neq 0$, then C contains a vertex of weight 1.

Proof. We first prove a parity estimate for inverse Gram matrices. We claim that, under the hypotheses of the lemma,

$$0 < \alpha \leq 1$$

and that every vector $\lambda \in \mathbb{Z}^C$ satisfying

$$\lambda_r \equiv 1 \pmod 2, \qquad \lambda_u \equiv 0 \pmod 2 \quad (u \neq r)$$

also satisfies

$$\lambda^T G \lambda \geq \alpha.$$

This is proved by induction on $|C|$.

If $C = \{r\}$, then $A = [w(r)]$. The root hypothesis gives $w(r) \geq 1$, and positive definiteness also gives $w(r) > 0$. Thus

$$\alpha = \frac{1}{w(r)} \leq 1.$$

If $\lambda_r = s$ is odd, then $s^2 \geq 1$, so

$$\lambda^T G \lambda = \frac{s^2}{w(r)} \geq \frac{1}{w(r)} = \alpha.$$

This proves the base case.

Assume now that $|C| > 1$. Let $C_1, \dots, C_m$ be the connected components of $C - \{r\}$, and let r_i be the unique vertex of C_i adjacent to r. Root C_i at r_i. Let A_i be the Gram matrix of C_i, set $G_i = A_i^{-1}$, let $e_i = e_{r_i}$, and put

$$\alpha_i = e_i^T G_i e_i.$$

Each A_i is positive definite because it is a principal submatrix of A. Also, each rooted tree C_i satisfies the same type of weight hypotheses: since $r_i \neq r$,

$$w(r_i) \geq d_C(r_i) = d_{C_i}(r_i) + 1,$$

and every other vertex $u \in C_i$ satisfies

$$w(u) \geq d_C(u) = d_{C_i}(u).$$

Therefore, by induction,

$$0 < \alpha_i \leq 1 \qquad \text{for all } i.$$

Order the vertices of C by putting r first and then the vertices of $C_1, \dots, C_m$. With respect to this order,

$$A = \begin{pmatrix} w(r) & -\eta^T \\ -\eta & B \end{pmatrix}, \qquad B = A_1 \oplus \cdots \oplus A_m,$$

where the C_i-block of η is e_i. By Lemma 3, if

$$S = w(r) - \eta^T B^{-1} \eta,$$

then $S > 0$ and

$$\alpha = (A^{-1})_{rr} = \frac{1}{S}.$$

Since $B^{-1} = G_1 \oplus \cdots \oplus G_m$, we have

$$\eta^T B^{-1} \eta = \sum_i \alpha_i.$$

Also $m = d_C(r)$, so the root hypothesis and the inductive bounds $\alpha_i \leq 1$ imply

$$S = w(r) - \sum_i \alpha_i \geq (m+1) - m = 1.$$

Thus $0 < \alpha = 1/S \leq 1$.

Now let $\lambda \in \mathbb{Z}^C$ have odd root coordinate and even nonroot coordinates. In the above ordering, write

$$\lambda = (s, 2\mu_1, \dots, 2\mu_m),$$

where s is odd and $\mu_i \in \mathbb{Z}^{C_i}$. Define

$$\rho_i = e_i^T G_i \mu_i, \qquad E_i = \mu_i^T G_i \mu_i, \qquad \rho = \sum_i \rho_i.$$

Using the inverse formula from Lemma 3 with $\gamma = -\eta$, we obtain

$$\lambda^T G \lambda = 4 \sum_i E_i + \frac{(s + 2\rho)^2}{S}.$$

For each i, the vectors $2\mu_i - e_i$ and $2\mu_i + e_i$ have odd coordinate at the root r_i and even coordinates elsewhere in C_i. The induction hypothesis applied to C_i gives

$$(2\mu_i - e_i)^T G_i (2\mu_i - e_i) \geq \alpha_i, \qquad (2\mu_i + e_i)^T G_i (2\mu_i + e_i) \geq \alpha_i.$$

Expanding these two inequalities gives

$$4E_i - 4\rho_i + \alpha_i \geq \alpha_i, \qquad 4E_i + 4\rho_i + \alpha_i \geq \alpha_i.$$

Hence

$$E_i \geq |\rho_i|.$$

Therefore

$$\lambda^T G \lambda \geq 4 \sum_i |\rho_i| + \frac{(s + 2\rho)^2}{S} \geq 4|\rho| + \frac{(s + 2\rho)^2}{S}.$$

Set $y = 2\rho$. Then

$$\lambda^T G \lambda \geq 2|y| + \frac{(s + y)^2}{S}.$$

Since $S \geq 1$, it is enough to prove

$$(s + y)^2 + 2S|y| \geq 1.$$

Let $a = |s| \geq 1$ and $u = |y|$. If $sy \geq 0$, then $|s + y| = a + u \geq 1$, so the inequality is immediate. If $sy < 0$, then

$$(s + y)^2 + 2S|y| \geq (a - u)^2 + 2u = \big(u - (a - 1)\big)^2 + 2a - 1 \geq 1.$$

Thus $S\lambda^T G \lambda \geq 1$, and hence

$$\lambda^T G \lambda \geq \frac{1}{S} = \alpha.$$

This completes the induction and proves the parity estimate.

We now derive the stated branch inequality. Fix $x \in L(C)$, let $c = x_r$, and set $Q = x^2 = x^T Ax$. For $k \in \mathbb{Z}$, put

$$t = 2k - 1.$$

Define

$$\lambda = 2Ax - te_r.$$

Because A has integer entries and x has integer coordinates, $\lambda \in \mathbb{Z}^C$. Its root coordinate is even minus the odd integer t, hence is odd, while every nonroot coordinate is even. The parity estimate therefore gives

$$\lambda^T G\lambda \geq \alpha.$$

Using $G = A^{-1}$, we expand:

$$\lambda^T G\lambda = (2Ax - te_r)^T G(2Ax - te_r) = 4Q - 4tc + t^2\alpha.$$

Hence

$$4Q - 4tc + t^2\alpha \geq \alpha,$$

or equivalently

$$tc - Q \leq \frac{t^2 - 1}{4}\alpha.$$

Since $t = 2k - 1$,

$$\frac{t^2 - 1}{4} = \frac{(2k-1)^2 - 1}{4} = k(k-1).$$

Therefore

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

It remains to prove the equality statement for $k = 0, 1$. For every $x \in L(C)$, expanding the form gives

$$x^2 = \sum_{\{u,v\}\in E(C)} (x_u - x_v)^2 + \sum_{u\in C} \big(w(u) - d_C(u)\big)x_u^2.$$

Indeed, in the first sum, the term x_u^2 appears once for each edge incident to u, hence $d_C(u)$ times, and each edge contributes the cross term $-2x_u x_v$. Therefore the displayed expression equals

$$\sum_{u\in C} w(u)x_u^2 - 2 \sum_{\{u,v\}\in E(C)} x_u x_v = x^2.$$

By the hypotheses,

$$w(r) - d_C(r) \geq 1, \qquad w(u) - d_C(u) \geq 0 \quad (u \neq r),$$

so

$$x^2 - c^2 = \sum_{\{u,v\}\in E(C)} (x_u - x_v)^2 + \big(w(r) - d_C(r) - 1\big)c^2 + \sum_{u\neq r} \big(w(u) - d_C(u)\big)x_u^2 \geq 0.$$

Thus $x^2 \geq c^2$.

Suppose first that $k = 1$, equality holds in the branch inequality, and $x \neq 0$. Since $k(k-1) = 0$, equality says

$$c - x^2 = 0.$$

Thus $x^2 = c$. Positive definiteness gives $x^2 > 0$, so $c > 0$, and hence $x^2 = |c|$. Since $x^2 \geq c^2$, we get $|c| \geq c^2$. Because $c \in \mathbb{Z}$ and $c \neq 0$, this forces $c = 1$, and then $x^2 = c^2 = 1$.

Similarly, if $k = 0$, equality says

$$-c - x^2 = 0.$$

Thus $x^2 = -c > 0$, so $c < 0$. Again $x^2 = |c|$, and the inequality $x^2 \geq c^2$ forces $c = -1$ and $x^2 = c^2 = 1$.

In either case, $|c| = 1$ and $x^2 = c^2$. Therefore equality holds in the nonnegative sum for $x^2 - c^2$, so every summand in that sum is zero. In particular, each edge term $(x_u - x_v)^2$ is zero. Since C is connected, all coordinates of x are equal to $c = \pm 1$.

If C has only one vertex, then $C = \{r\}$. The equality $x^2 = c^2$, with $c^2 = 1$, gives

$$(w(r) - d_C(r) - 1)c^2 = 0.$$

Since $d_C(r) = 0$, we obtain $w(r) = 1$.

If C has more than one vertex, choose a vertex $u \neq r$ at maximal distance from r. Such a vertex is a nonroot leaf, so $d_C(u) = 1$. Because all coordinates of x are ± 1, we have $x_u^2 = 1$. The vanishing of the corresponding nonroot summand gives

$$\big(w(u) - d_C(u)\big)x_u^2 = 0,$$

hence $w(u) - d_C(u) = 0$, and therefore $w(u) = 1$. Thus, whenever equality holds for $k = 0$ or $k = 1$ with $x \neq 0$, the branch C contains a vertex of weight 1. $\square$

Lemma 5 (Calculation at a vertex)**.** *Let T be a finite weighted tree whose form is positive definite, and let b be a vertex of T. Let C range over the components of $T - \{b\}$. For each such component C, let r_C be the unique vertex of C adjacent to b, let A_C be the Gram matrix of C, and define*

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Then

$$\sum_C \alpha_C < w(b).$$

Moreover, if

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$ and $x_C \in L(C)$, and if $c_C = (x_C)_{r_C}$, then

$$x \cdot (b - x) = k(1-k)w(b) + \sum_C \Big((2k-1)c_C - x_C^2 \Big).$$

Proof. If $T = \{b\}$, then there are no components C, the sum $\sum_C \alpha_C$ is empty, and the first assertion says $0 < w(b)$, which follows from positive definiteness because $w(b) = b^2$. The displayed formula also reduces immediately to

$$(kb) \cdot (b - kb) = k(1-k)w(b).$$

Thus assume $T \neq \{b\}$.

Order the vertices of T by putting b first, followed by the vertices in the components C of $T - \{b\}$. With respect to this order, the full Gram matrix has block form

$$\begin{pmatrix} w(b) & -\eta^T \\ -\eta & B \end{pmatrix}, \qquad B = \bigoplus_C A_C,$$

where the C-block of η is e_{r_C}. The matrix B is positive definite because it is a principal submatrix of the positive definite Gram matrix of T. Applying Lemma 3, the Schur complement is positive:

$$w(b) - \eta^T B^{-1} \eta > 0.$$

Since $B^{-1} = \bigoplus_C A_C^{-1}$, we have

$$\eta^T B^{-1} \eta = \sum_C \alpha_C.$$

Therefore

$$\sum_C \alpha_C < w(b).$$

Now write

$$x = kb + \sum_C x_C.$$

Different components of $T - \{b\}$ are orthogonal to one another, and b pairs nontrivially with a component C only at the root r_C, where $b \cdot r_C = -1$. Hence

$$b \cdot x = kw(b) - \sum_C c_C.$$

Also,

$$x^2 = k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2.$$

Therefore

$$\begin{aligned} x \cdot (b - x) &= x \cdot b - x^2 \\ &= \left(kw(b) - \sum_C c_C \right) - \left(k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2 \right) \\ &= k(1-k)w(b) + \sum_C \Big((2k-1)c_C - x_C^2 \Big). \end{aligned}$$

This proves the formula. $\square$

Proof of the theorem

Proof of Theorem 1. Let b denote the unique vertex of T satisfying $w(b) < d_T(b)$. Thus every vertex $u \neq b$ satisfies

$$w(u) \geq d_T(u).$$

If T contains a vertex of weight 1, then Lemma 2 shows that this vertex is irreducible, and the theorem follows. Hence assume from now on that T contains no vertex of weight 1. We will prove that b is irreducible.

Suppose, for contradiction, that b is reducible. Then there exists $x \in L(T)$ such that

$$x \neq 0, \qquad b - x \neq 0, \qquad x \cdot (b - x) \geq 0.$$

Decompose x according to the components of $T - \{b\}$:

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$, each $x_C \in L(C)$, and the sum ranges over the components C of $T - \{b\}$. Let r_C be the vertex of C adjacent to b, and put

$$c_C = (x_C)_{r_C}.$$

We now verify that each rooted component C, rooted at r_C, satisfies the hypotheses of Lemma 4. Since $r_C \neq b$, the uniqueness of b gives

$$w(r_C) \geq d_T(r_C) = d_C(r_C) + 1,$$

because the only edge incident to r_C that is not in C is the edge from r_C to b. If $u \in C$ and $u \neq r_C$, then no edge incident to u was removed when passing from T to C, so

$$w(u) \geq d_T(u) = d_C(u).$$

Finally, the Gram matrix A_C of C is positive definite because it is a principal submatrix of the positive definite Gram matrix of T. Thus Lemma 4 applies to each component C.

Let

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Applying Lemma 4 to each component, with the same integer k, gives

$$(2k - 1)c_C - x_C^2 \leq k(k - 1)\alpha_C.$$

Using Lemma 5, we obtain

$$\begin{aligned} x \cdot (b - x) &= k(1 - k)w(b) + \sum_C \Big((2k - 1)c_C - x_C^2\Big) \\ &\leq k(1 - k)w(b) + k(k - 1)\sum_C \alpha_C \\ &= k(k - 1)\left(\sum_C \alpha_C - w(b)\right). \end{aligned}$$

By Lemma 5,

$$\sum_C \alpha_C - w(b) < 0.$$

If $k \notin \{0,1\}$, then $k(k-1) > 0$, so the last upper bound is strictly negative. This contradicts $x \cdot (b-x) \geq 0$. Therefore

$$k \in \{0,1\}.$$

For $k = 0$ or $k = 1$, we have $k(1-k) = 0$, and Lemma 4 gives

$$(2k-1)c_C - x_C^2 \leq 0 \qquad \text{for every component } C.$$

Lemma 5 therefore gives

$$x \cdot (b-x) = \sum_C \Big((2k-1)c_C - x_C^2 \Big),$$

a sum of nonpositive terms. Since $x \cdot (b-x) \geq 0$, every term in this sum must be equal to 0. Hence, for every component C,

$$(2k-1)c_C - x_C^2 = 0.$$

If some $x_C \neq 0$, then the equality statement in Lemma 4 implies that C, and therefore T, contains a vertex of weight 1. This contradicts our standing assumption that T has no weight-one vertex. Thus

$$x_C = 0 \qquad \text{for every component } C.$$

Consequently $x = kb$. If $k = 0$, then $x = 0$, contradicting $x \neq 0$. If $k = 1$, then $b - x = 0$, contradicting $b - x \neq 0$. Both possibilities are impossible.

Therefore no such decomposition of b exists, and b is irreducible. Hence T contains an irreducible vertex. □